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EP1108 Modern Physics - Exam 1

Please fill in your name and roll number. Multiple choices may be correct, credit will be given only if ALL the correct answers are chosen and none of the wrong ones. Total time of the exam is one hour. You have to use your IITH email ID and you will be allowed to submit your response only once. Take screenshots of your responses as a backup before submitting.

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The results of the Michelson-Morley experiment were incompatible with the axioms of which of the following theory(ies)? [1 pt]

- [x] Galilean Relativity/Mechanics
- [ ] Special Relativity
- [ ] General Relativity
- [ ] None of the above

Which of the following quantity/quantities is conserved after a Lorentz transformation? (vx,vy,vz are x,y,z components of velocity) [1 pt]

- [ ] x^2
- [x] x^2+y^2+z^2-c^2t^2
- [x] c^2-vx^2-vy^2-vz^2
- [ ] None of the above

Ratio of emitted powers from two stars which behave as black bodies is equal to 16:1. What is the ratio of their temperatures [1 pt]

- [ ] 16:1
- [x] 2:1
- [ ] 1:16
- [ ] 1:2

A relativistic particle of mass m and Kinetic energy 2mc^2 strikes and sticks to a stationary particle of rest mass 2m. What is the rest mass of the composite particle after the collision? [3 pts]

- [ ] 5m
- [ ] 4m
- [ ] sqrt(17) m

- [x] sqrt (13) m

An electron has a Kinetic energy of 0.3 MeV. If its rest mass is equal to 0.5 MeV/c^2, what is its velocity according to Special Relativity? [1 pt]

- [x] 0.78 c
- [ ] 0.61 c
- [ ] 0.5 c
- [ ] 0.95 c

A muon produced in the Earth's atmosphere has an energy of 100 GeV. If its rest mass is equal to 100 MeV and lifetime in rest frame equal to 2 microseconds, what will be its lifetime as seen by an observer on the Earth? (1 GeV=1000 MeV) [1 pt]

- [ ] 2 microseconds
- [x] 2000 microseconds
- [ ] 200 microseconds
- [ ] None of the above

Assume a spaceship has a speed of 0.9c with respect to the Earth. If we want to pass by the spaceship with a relative speed (with respect to the ship) of 0.5c, what should be our relative speed with respect to the Earth. [2 pts]

- [ ] 1.4c
- [ ] 0.4c
- [x] 0.965c
- [ ] None of the above

The work function of Sodium is 2.3 eV. What is the maximum wavelength of light that will cause photoelectrons to be emitted from Sodium? (1 eV =1.6 x 10^{-19} J; c=3 x 10^8 m/s ; h=6.634 x 10^{-34} m^2 kg/sec ; 1A = 10^{-10} m) [2 pts]

- [ ] 1000 A
- [ ] 2300 A
- [x] 5380 A
- [ ] 100 A

Which of the following features of blackbody radiation could not be explained by the classical theory (or Rayleigh-Jeans theory)? [1 pt]

- [ ] Wein's law
- [ ] long-wavelength limit
- [x] short-wavelength limit
- [ ] None of the above

From measuring the spectrum of two stars A and B (both of whom are black-body emitters), if we find spectrum of star A is peaked at longer wavelengths and that of star B is peaked at shorter wavelengths. Which of the stars will have a higher temperature? [1 pt]

- [ ] Star A
- [x] Star B
- [ ] Both will have same temperature
- [ ] Cannot be determined.

Which of the following is not true considering the Classical Harmonic Oscillator (CHO) and the Quantum Harmonic Oscillator (QHO)? [1 pt]

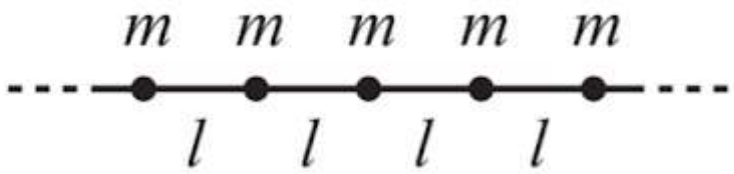
- [ ] The probability distribution for finding CHO is similar to that of QHO for a large value of quantum number, n
- [ ] At a very high temperature, the ground state energies of the CHO and the QHO are same
- [ ] Resistive force like the frictional force due to air is important for CHO, not for QHO

- [x] There exist a "forbidden" or "impossible to penetrate" region in CHO but no such finite region exists in QHO
- [ ] None of the above

Which of the following cannot be explained by Heisenberg's uncertainty principle (HUP)? [1 pt]

- [ ] In case of the plane parallel light passing through a slit or a tiny hole, tinier the hole, more circular and symmetric the emergent wavefront
- [ ] Naturally broad spectral lines imply that the lifetime of an electron in a given energy state is finite
- [ ] For a given microscope, higher energy photons are better probe to determine the position of an electron more accurately
- [x] The higher the kinetic energy of two positively charged colliding particles, the higher the probability they will fuse together
- [ ] None of the above

Consider a system that is made up of beads on a massless string. The beads have mass m and are glued to the string with separation, l, shown in Figure. The tension is T. The dispersion relation is given below. Find out the relation between the group velocity (Vg) and phase velocity (Vp) if k*l << 1 [2 pt]



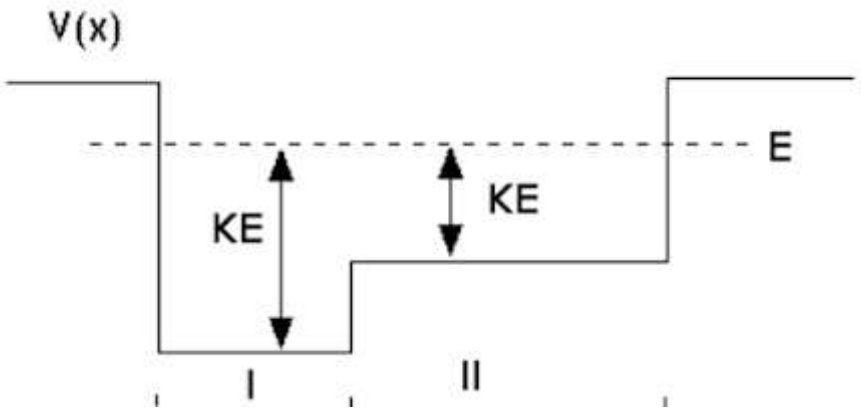
$$\omega(k) = 2\omega_0 \sin(k\ell/2), \text{ where } \omega_0 \equiv \sqrt{T/m\ell}.$$

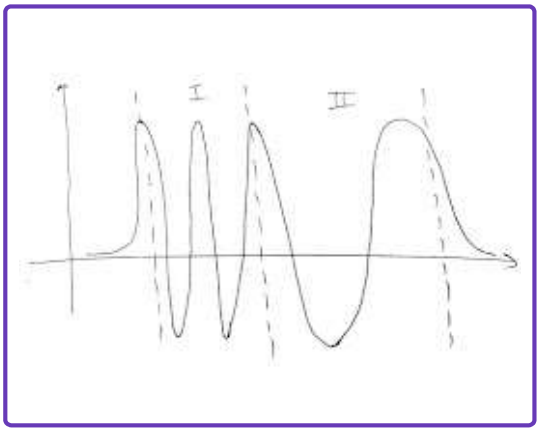
- [x] Vp = Vg
- [ ] Vp = 2*Vg
- [ ] Vp = T*Vg
- [ ] None of the above

For a 1 cm radio wave, if the slit separation is 1 cm, the distance of the screen from the slit is 10 cm, at what distance on the screen the radio intensity of fringes will fall to 65% of the maximum intensity? [2 pt]

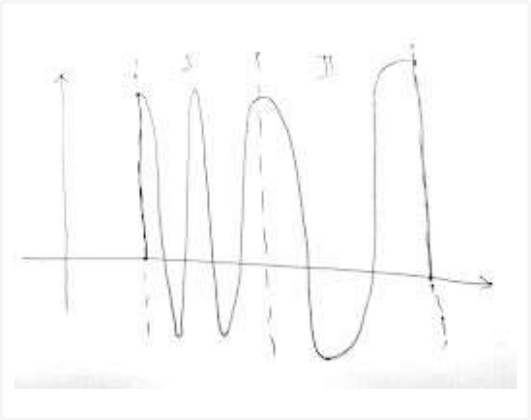
- [ ] ~6 mm
- [x] ~2 cm
- [ ] ~11 cm
- [ ] ~1.3 km
- [ ] ~0.00024 mm
- [ ] None of the above

A particle of the total energy E is kept in the potential barrier V(x), shown below. What would be the correct representation of the wave function of the particle in different regions? Vertical dotted lines in the options represent the boundaries of region I and II [1 pt]

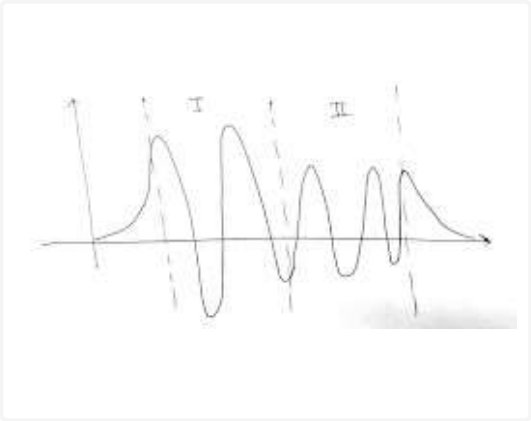




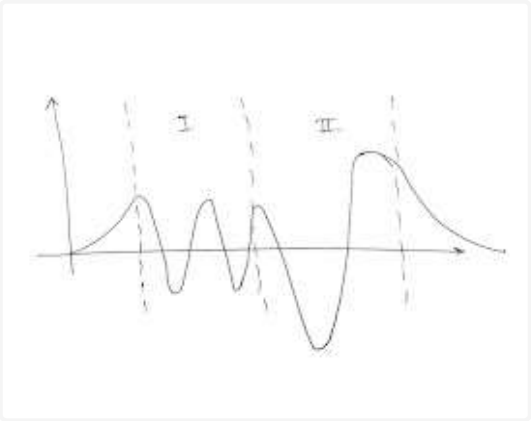
☑ Option 1



☐ Option 2



☐ Option 3



☐ Option 4

☐ None of the above

An operator acted upon an eigenfunction returns the eigenvalue of hc/λ. What is the differential form of the operator? [1 pt]

☐ +i*h-bar*d/dx

☑ -i*h-bar*d/dx

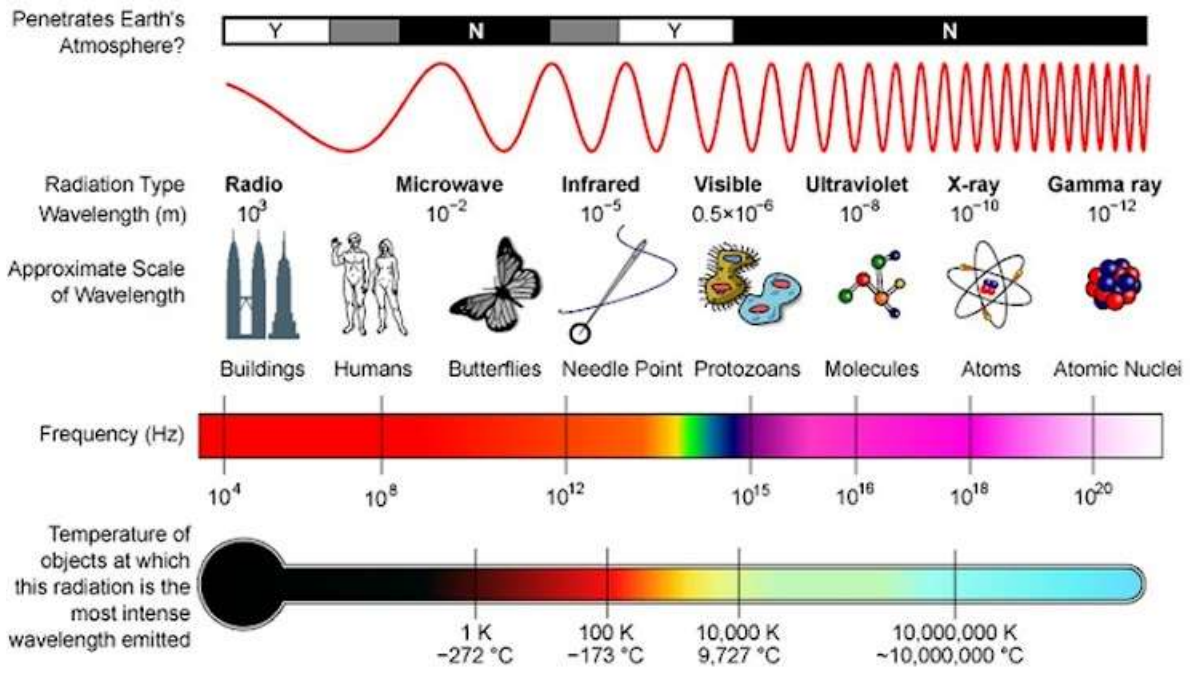
☐ +i*h-bar*d/dt

- [ ] -i*h-bar*d/dt
- [ ] None of the above

If the width of a barrier (L) in a potential well is 5 times then the barrier penetration depth of the particle (represented by 1/α, where α = sqrt(2m(U-E))/h-bar), what the reflection coefficient of the wavefunction at the barrier? [2 pt]

- [ ] 0.0067
- [x] 0.9932
- [ ] 0.67
- [ ] 0.00099
- [ ] 0.067
- [ ] None of the bove

Let's consider a hypothetical particle is kept in an infinite square well potential of length, α. If m is the mass of the particle, then let's assume π^2*h-bar^2/(2*m*α^2) = 1 eV. If the particle makes a transition from the 25th quantum state to 10th quantum state, the energy released will be in the form of a quantized radiation. What type of radiation detector is required to record such an event? 1 eV = 2.41 x 10^14 Hz [2 pt]



- [ ] An Infrared Detector
- [ ] An Optical Detector
- [ ] An Ultraviolet Detector
- [x] An X-ray Detector
- [ ] No radiation detector can detect such photon since its energy would be very low

Which of the following cannot be described by the quantum mechanical wavefunction? [1 pt]

- [ ] Superconductivity
- [ ] Atomic structure of solids and gases
- [ ] Lasers
- [ ] Properties of elementary particles
- [x] None of the above

An "excited" atom gets rid of its excess energy by emitting one or more photons of appropriate frequency. The mean life time of the excited state, i.e., the average time that elapses between the excitation of an atom and the emission of radiation is 10^-8 sec. What is the limit imposed by the uncertainty principle on the accuracy with which the frequency of the radiation emitted by an atom can be determined? [2 pt]

- [x] ~1-2 x 10^7 Hz
- [ ] ~ 3-4 x 10^5 Hz
- [ ] ~2 x 10^14 Hz
- [ ] ~500-1094 Hz
- [ ] None of the above
- [ ] Cannot be determined

What is the total energy of a photon of momentum p and frequency f? [1 pt]

- [x] pc
- [ ] sqrt(p^2 c^2+ h^2 f^2)
- [ ] 0
- [ ] None of the above

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